

Properties of materials

SUBSTANCES CAN BE UNDERSTOOD BY OBSERVING THEIR PROPERTIES.

SEE ALSO

Periodic table	116–117 >
Corrosion	133 >
What is mass?	172 >
Stretching and deforming	174–175 >

Every substance has its own unique set of properties—color, density, smell, and flammability. Chemists try to understand why the substances in nature have such varied properties.

Mass and density

All objects have a mass: a measure of how much matter they contain. Mass is not an indicator of size. A piece of lead has more mass than an identically sized piece of polystyrene, for example. The difference in mass is due to a property known as density: a measure of how tightly packed matter is inside a substance. Density is calculated as mass divided by volume, and expressed with the units kg/m^3 —or often g/cm^3 . Lead is one of the most dense elements of all, which is why it is used in weights—a small, manageable lead object contains a lot of matter and so weighs a large amount.

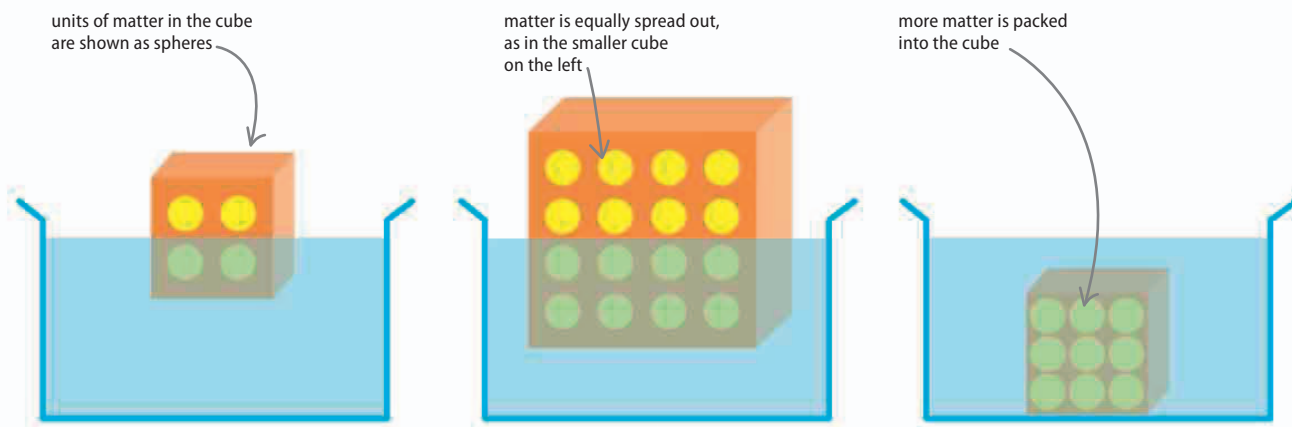
Buoyancy

The density of a substance can be tested by putting it in water. If an object has a density higher than water, it will sink; if it is less dense than water, it will float.

REAL WORLD

Physical versus chemical

The spokes of this bike are bent. The bending is due to the physical properties of the metal. Physical properties do not change the substance (in this case, metal). Some parts of the bike have rusted. The rusting is due to the chemical properties of the metal. Chemical properties relate to how the substance changes into other materials (rust) when it reacts with other substances (air and water).



△ Low-density object

This cube is less dense than water. The matter in the cube is spread out more than the matter in water, so it weighs less than an equal volume of water.

△ Larger object

This cube is made of the same material as the first, only it has four times the volume—and weighs four times as much. So it has the same density as the first and floats.

△ High-density object

This cube is the same size as the first cube, but has a higher density. In this case, the cube weighs more than the same volume of water, so it sinks.